

Multiple Choice Questions (MCQs) on Virtual Memory

1. What is the primary purpose of virtual memory?

- A) To increase the physical RAM
- B) To provide the illusion of more memory than physically available
- C) To speed up the CPU
- D) To store data permanently
- **Correct Answer: B**

2. Which of the following is a key feature of virtual memory?

- A) It allows programs to run without any memory
- B) It provides memory isolation between processes
- C) It eliminates the need for secondary storage
- D) It requires more physical memory
- **Correct Answer: B**

3. In demand paging, when is a page loaded into memory?

- A) When the program starts
- B) When the entire program is loaded
- C) Only when it is required
- D) At regular intervals
- **Correct Answer: C**

4. What occurs during a page fault?

- A) The program runs faster
- B) The operating system pauses the process
- C) The page is deleted from memory
- D) The CPU is reset
- **Correct Answer: B**

5. Which type of page fault requires loading a page from disk?

- A) Major Page Fault
- B) Minor Page Fault
- C) Soft Page Fault
- D) Hard Page Fault
- **Correct Answer: A**

6. What does the FIFO page replacement algorithm do?

- A) Replaces the most recently used page
- B) Replaces the oldest page in memory
- C) Replaces the page that will be used soonest
- D) Replaces pages randomly
- **Correct Answer: B**

7. Which page replacement algorithm is considered theoretically the best?

- A) FIFO
- B) LRU
- C) Optimal Page Replacement
- D) Clock Algorithm

- **Correct Answer: C**

8. What is a disadvantage of the Least Recently Used (LRU) algorithm?

- A) It is too simple to implement
- B) It may replace frequently used pages
- C) It requires tracking access times
- D) It does not reduce page faults

- **Correct Answer: C**

9. In the Clock Algorithm, what does a reference bit indicate?

- A) The page is not needed
- B) The page has been accessed recently
- C) The page is the oldest
- D) The page is currently in use

- **Correct Answer: B**

10. If a program requires 10 MB of memory but only 2 MB of pages are active, what technique is being used?

- A) Paging
- B) Demand Paging
- C) Swapping
- D) Segmentation

- **Correct Answer: B**

11. What happens when a page is replaced using the Optimal Page Replacement algorithm?

- A) The page that has been in memory the longest is replaced
- B) The page that will not be used for the longest time in the future is replaced
- C) The page that is least recently used is replaced
- D) The page is replaced randomly

- **Correct Answer: B**

12. Which of the following statements about virtual memory is true?

- A) It eliminates the need for physical memory
- B) It allows multiple processes to share the same memory space
- C) It can lead to increased disk I/O
- D) It is only used in high-performance computing

- **Correct Answer: C**